AVIATION INCIDENT ANALYSIS

A DATA STORY

Exploring Patterns, Unusual Observations and Trends in Aviation Incidents (2008 – 2025)

Data
Patterns
Trends
Insights
Safer Skies

1. SETTING THE SCENE

Aviation connects people and places, but incidents continue to raise important questions about safety, environmental conditions, aircraft characteristics and human performance. We use data analysis to uncover patterns and trends in aviation incidents.

2. OUR DATA

29,708 records

50 original columns

2008 – 2025 Aviation incident data

3. OBJECTIVES

- Clean and prepare the dataset
- Remove redundant and low-variance features
- Apply PCA, SVD, t-SNE and LLE
- Analyze trends and seasonal patterns in incidents
- Visualize results and identify key findings

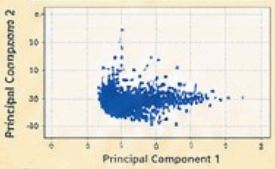
4. DATA PREPARATION

- Removed highly correlated feature (inj_tot_t)
- Handled missing values (median imputation)
- Removed low-variance features (threshold = 0.01)
- Numerical features: 21 → 19

5. DIMENSIONALITY REDUCTION

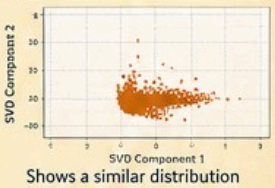
Visualizing high-dimensional data in 2D

PCA



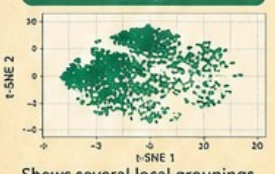
First two components explain 17.89% of variance. Most points are concentrated near the centre, with a few outliers.

SVD



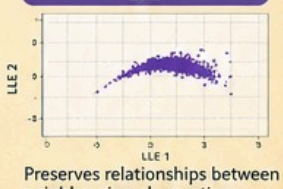
Shows a similar distribution to PCA, with a dense central group and a few distant observations.

t-SNE



Shows several local groupings of observations, indicating possible similarities between records.

LLE



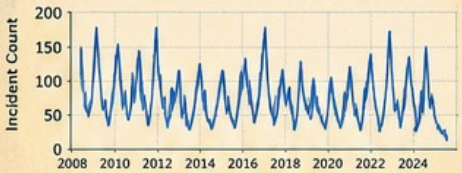
Preserves relationships between neighbouring observations, showing a dense cluster with some scattered points.

DIFFERENT PERSPECTIVES
SAME STORY
THE BIGGER PICTURE

6. TIME-SERIES ANALYSIS

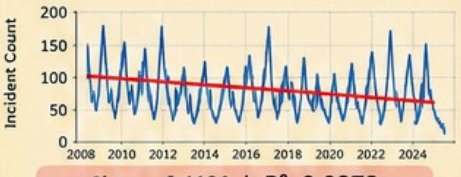
How aviation incident counts changed over time

Monthly Incident Counts



Repeated peaks and drops, indicating considerable variation in monthly incident counts.

Linear Regression Trend



Slope: -0.1101 | R^2: 0.0273

A slight downward trend, but the low R^2 value (2.73%) shows that the data fluctuate too much for a straight line to describe them well.

Incidents by Month



Shows a seasonal pattern, with higher counts in some months (e.g., July) and lower counts in others.

7. KEY FINDINGS

1. Dataset contains 29,708 aviation incident records (2008–2025).
2. Numerical features reduced from 21 to 19.
3. PCA first two components explain 17.89% of variance.
4. t-SNE and LLE reveal clustered patterns.
5. Monthly incident counts show considerable variation.
6. Linear trend has a slight downward slope (−0.1101) with low R^2 (0.0273).
7. July has the highest aggregate incident count.

8. THE BIG PICTURE

Patterns Today
Safer Skies Tomorrow

- Most incidents occur under clear sky conditions and involve relatively few injuries.
- Dimensionality reduction helps visualize patterns and unusual observations.
- Incident counts fluctuate with seasonal variation.
- A simple linear model explains only a small portion of the variation.
- Aviation incidents are complex, with many interacting factors.

CASE STUDY 2: AVIATION ACCIDENT ANALYSIS

Exploratory Data Analysis, Dimensionality Reduction and Time-Series Analysis

1. Introduction

Exploratory Data Analysis (EDA) helps us understand a dataset by identifying patterns, missing values, relationships, and unusual observations.

In this case study, an aviation incident dataset was analyzed using Python. Feature selection, dimensionality reduction, and time-series analysis were performed to understand the data and study how incident counts changed over time.

Objectives

- Clean and prepare the dataset.
- Remove redundant and low-variance features.
- Apply PCA, SVD, t-SNE, and explore LLE.
- Analyze trends and seasonal patterns in aviation incidents.
- Visualize the results and identify important findings.

2. Dataset and Preprocessing

The dataset contained **29,708 records and 50 columns**, covering aviation incidents from 2008 to 2025.

2.1 Correlation-Based Feature Selection

Correlation analysis was used to identify highly related numerical features. A threshold of 0.90 was used, and inj_tot_t was identified for removal.

After removing this column, the dataset contained 49 columns.

```
num = df.select_dtypes(include=np.number)

corr = num.corr().abs()

upper = corr.where(
    np.triu(np.ones(corr.shape), k=1).astype(bool)
)

to_drop = [
    col for col in upper.columns
    if any(upper[col] > 0.90)
]

print("Features to remove:", to_drop)

df_filter = df.drop(columns=to_drop)
print(df_filter.shape)
```

```
Features to remove: ['inj_tot_t']
(29708, 49)
```

Figure 1: Removal of a highly correlated feature.

2.2 Missing Values and Feature Selection

Numerical columns containing only missing values were removed. Remaining missing values were filled using the median. The VarianceThreshold method with a threshold of 0.01 was then used to remove low-variance features.

The number of numerical features decreased from 21 to 19.

Original numerical features: 21

Selected features: 19

Selected columns: ['sky_ceil_ht', 'vis_sm', 'wind_dir_deg', 'wind_vel_kts', 'gust_kts', 'wx_temp', 'inj_tot_f', 'inj_tot_m', 'inj_tot_n', 'inj_tot_s', 'Aircraft_Key', 'num_eng', 'rwy_len', 'rwy_width', 'crew_age', 'phase_no', 'flight_hours', 'eng_time_total', 'eng_time_last_insp']

Reduced dataset shape: (29708, 19)

An additional feature-selection output showed different matrix dimensions.

```
print(type(X))
print("X shape:", X.shape)
print("RFE support length:", len(rfe.support_))
print("Selected feature names:", len(selected))
```

```
<class 'pandas.DataFrame'>
X shape: (29708, 23)
RFE support length: 21
Selected feature names: 19
```

Figure 2: Additional feature-selection results.

3. Dimensionality Reduction

Dimensionality reduction converts data with many features into fewer dimensions, making it easier to visualize patterns.

3.1 Principal Component Analysis (PCA)

PCA transforms the original features into new components that capture as much variation in the data as possible.

The first two components explained approximately **17.89% of the total variance**.

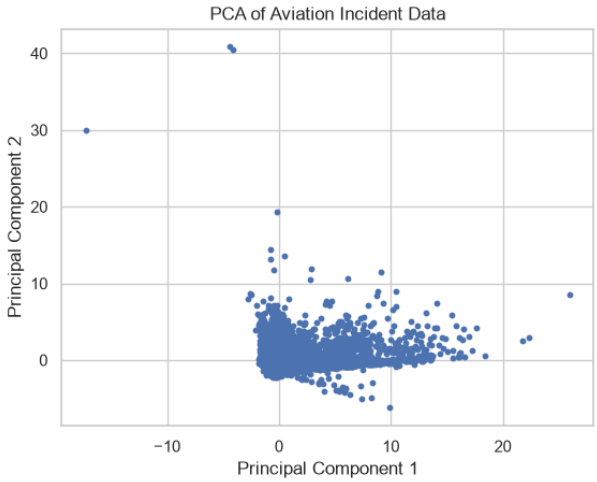


Figure 3: PCA visualization of aviation incidents.

Most observations are concentrated near the centre, while a few points appear farther away. These points may represent unusual observations. Since the first two components retain only a small portion of the total variance, the plot does not show the complete structure of the dataset.

3.2 Singular Value Decomposition (SVD)

SVD is a matrix-factorization technique that can reduce data into a smaller representation.

The two-dimensional SVD plot showed a distribution similar to PCA, with a dense central group and a few distant observations.

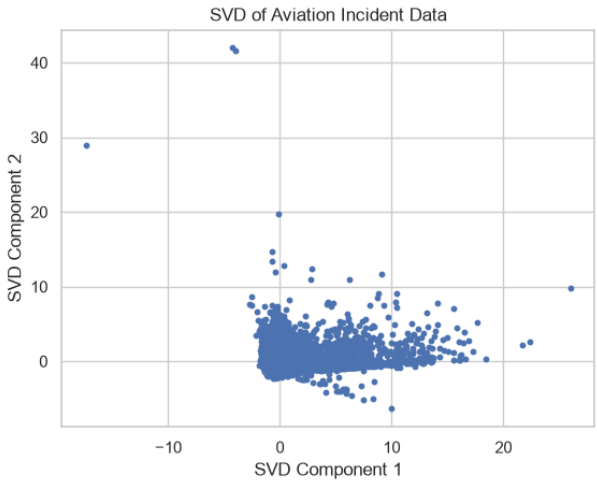


Figure 4: SVD visualization of aviation incidents.

3.3 t-SNE

t-SNE is a nonlinear technique that places similar observations close together in a lower-dimensional space.

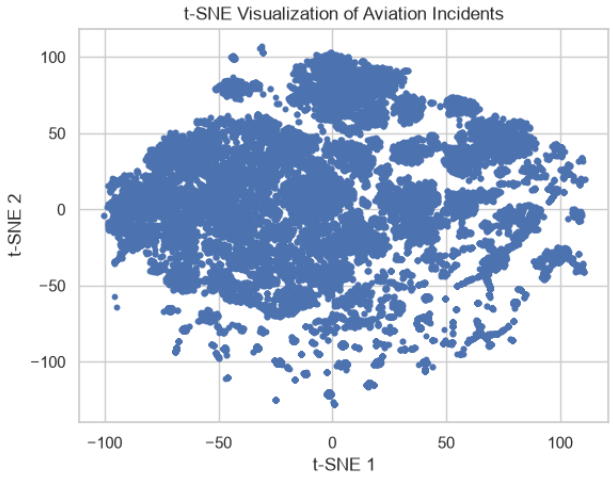


Figure 5: t-SNE visualization of aviation incidents.

The plot shows several local groupings of observations. These groups may indicate similarities between records, but further analysis is needed to understand what they represent.

3.4 Local Linear Embedding (LLE)

LLE is another nonlinear dimensionality-reduction technique. It tries to preserve relationships between neighbouring observations when reducing the number of dimensions.

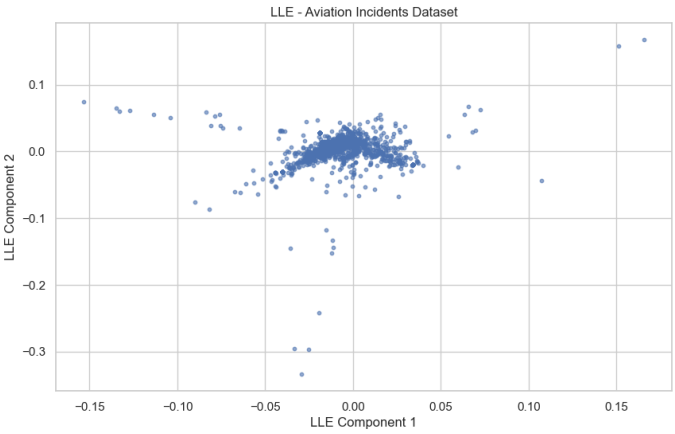


Figure 6: LLE Visualization of the Aviation Incidents Dataset.

4. Time-Series Analysis

Time-series analysis was used to study how aviation incident counts changed over time.

4.1 Date Conversion and Indexing

The ev_date column was initially stored as strings. It was converted into datetime format using Pandas. The records were then sorted by date, and the date column was set as the index.

4.2 Monthly Incident Counts

The records were grouped into monthly counts and displayed using a line chart.

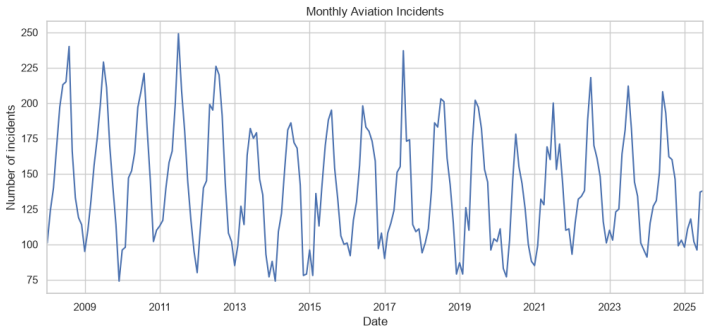


Figure 7: Monthly aviation incident counts over time.

The chart shows repeated peaks and drops, indicating considerable variation in monthly incident counts.

4.3 Linear Regression Trend

A linear regression line was fitted to identify the overall direction of change.

The results were:

- **Slope:** −0.1101
- **R-squared:** 0.0273

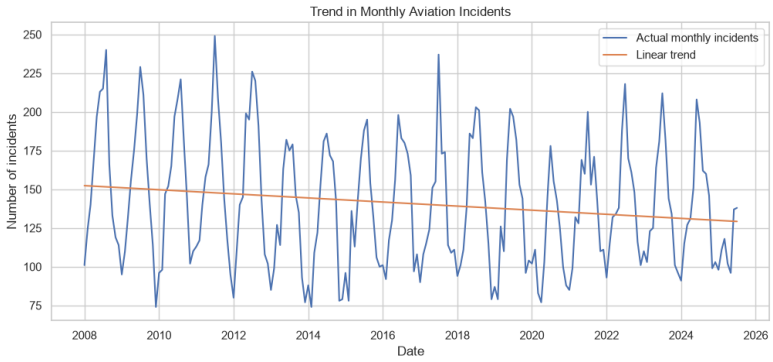


Figure 8: Linear trend in monthly aviation incidents.

The negative slope suggests a slight downward trend. However, the low R-squared value means that the linear model explains only about 2.73% of the variation. Therefore, the data fluctuate too much for a straight line to describe them well.

4.4 Incidents by Month

The records were grouped by calendar month to compare incident counts across the year.

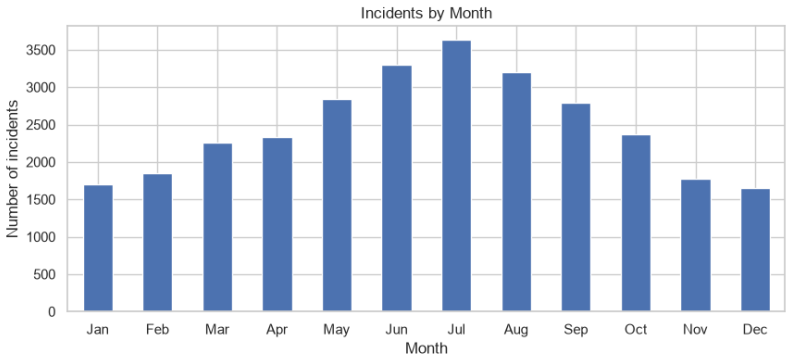


Figure 9: Distribution of incidents across calendar months.

July had the highest aggregate count, followed by June and August. November and December had lower counts in the displayed chart.

These are raw incident counts, not risk rates. More information about the number of flights would be needed to compare actual risk between months.

4.5 Year-Month Heatmap

A heatmap was created to compare incident counts across different years and months.

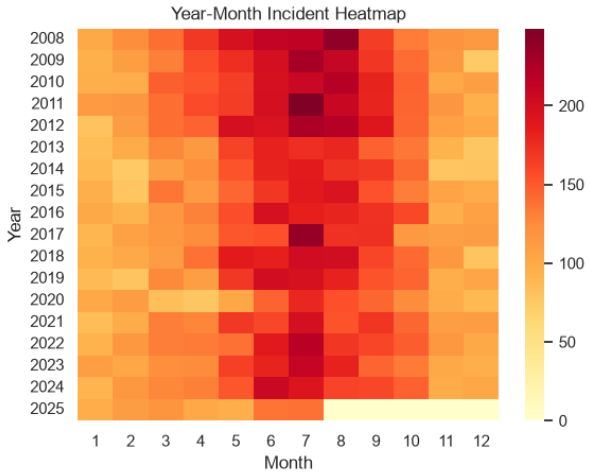


Figure 10: Year-month heatmap of aviation incidents.

Darker cells indicate higher counts, while lighter cells indicate lower counts. Higher counts appear frequently around June, July, and August. The 2025 data are incomplete, so the final row should be interpreted cautiously.

4.6 Seasonal Decomposition

Seasonal decomposition separates a time series into four parts:

- **Observed:** Original monthly counts.

- **Trend:** Long-term movement.
- **Seasonal:** Repeating patterns.
- **Residual:** Variation not explained by the trend and seasonal components.

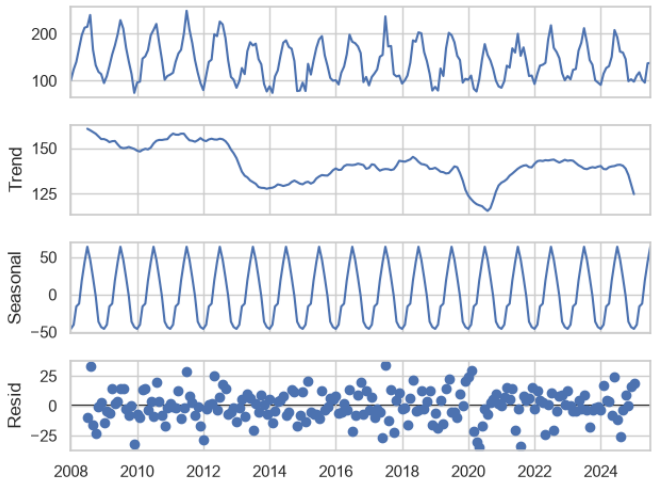


Figure 11: Seasonal decomposition of aviation incident counts.

The seasonal component shows a repeating annual pattern. The trend changes over time, with periods of decline and recovery. The residual component shows additional fluctuations not explained by the other components.

5. Key Findings

The main findings from the analysis are:

1. **Dataset size:** The original dataset contained 29,708 records and 50 columns.
2. **Feature selection:** Correlation screening removed inj_tot_t, while variance filtering reduced the numerical feature set from 21 to 19.
3. **PCA and SVD:** Both methods showed a dense group of observations and some distant points.
4. **PCA variance:** The first two components explained approximately 17.89% of the total variance.
5. **t-SNE:** The visualization showed local groupings of similar observations.
6. **Monthly patterns:** July had the highest aggregate incident count in the displayed chart.
7. **Trend:** The fitted linear trend was slightly negative but had low explanatory power.
8. **Seasonality:** The monthly charts and decomposition showed recurring patterns during the middle of the year.

6. Limitations

- Incident counts do not directly measure aviation risk because flight activity was not considered.
- The 2025 data cover only part of the year.
- PCA's first two components retain only 17.89% of the variance.
- The dimensionality-reduction plots do not identify the meaning of individual groups.
- The feature-selection outputs contain inconsistent matrix dimensions that need checking.
- The LLE result still needs to be verified.
- The observed patterns do not establish the causes of aviation incidents.

7. Conclusion and Future Work

This case study used EDA, feature selection, dimensionality reduction, and time-series analysis to explore an aviation incident dataset.

PCA, SVD, and t-SNE provided different ways to visualize the numerical data. The time-series analysis revealed substantial monthly variation, a weak downward linear trend, and recurring seasonal patterns.

Future work could include completing the LLE analysis, checking feature-selection consistency, investigating unusual observations, and comparing more advanced time-series models. Using flight counts to calculate incident rates would also help provide a more meaningful assessment of aviation safety.